Ching-Tsung (Deron) Tsai

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WORK EXPERIENCE

Amazon, New York, NY **Apr. 2025-Present**

Business Intelligence Engineer II, Dec. 2025-Present

Business Intelligence Engineer I, Apr. 2025-Nov. 2025

- Fine-tuned Qwen-2.5 via Direct Preference Optimization to reduce production RAG inference cost & latency by 21% while preserving response quality through consensus-based LLM-as-judge evaluation, achieving 98% satisfaction rate across 25,000+ queries
- Developed full-stack demo platform using Streamlit, DynamoDB, S3 to satisfy 100+ stakeholder requirements and align SVP-level decision-making on ads-marketing roadmap
- Built agentic pipeline (ReACT + self-reflection) to automate knowledge graph construction from 5,400+ webpages, serving as primary knowledge base for 5 research scientists across 2 workstreams

Pfizer (formerly Seagen), Seattle, WA (Remote) **Oct. 2023-Mar. 2025**

Senior Software Developer, Oct. 2024-Mar. 2025

Software Developer, Oct. 2023- Oct. 2024

- **Standardized hierarchical data model transformation** & quality check by developing an R package, actively used by 11 analysts
- Automated extraction of biomarker data and data-migration-related documents from 10+ sources using Python pandas, saving a 4-person development team **32 hours weekly** in building ETL pipelines
- Maintained **8** clinical DBs with >32,000 patient records via Apache Airflow and Docker on Azure. Enabled efficient downstream analysis

Mamon11, New York, NY (Remote) **Jul. 2023-Sep. 2023**

Data Science Intern

- Processed online transaction & user behavior data with PostgreSQL & Python polars, successfully streamlining ETL pipeline for **2** clients
- Boosted time-series XGBoost model performance via feature engineering, achieved **13.5% rise in clickthrough rate** & increase revenue

Regeneron, Tarrytown, NY **Jun. 2022-Aug. 2022**

Data Engineer Intern

- Transformed ~500G in-house genomic sequences into structured database with regex & NLP, enabling automatic feature engineering
- Built drug target identification pipeline on AWS DNAnexus, discovering **4** sequence patterns using multinomial regression

Chang Gung Memorial Hospital, Inst. of Stem Cell and Translational Cancer Research, New Taipei, Taiwan **Aug. 2019-Feb. 2021**

Junior Statistician

- **Designed evaluation metrics** for survival and Cox hazards model, discovering 3 factors to help identify appropriate surgical candidates

PROJECTS

LyricChat: Agentic RAG Chatbot for Emotion-Driven Song Recommendations (GitHub link) **Aug. 2024-Apr. 2025**

- Deployed empathetic chatbot with ReACT framework to recommend mood-matching songs, driving fan engagement for an online singer and placing in **top 10** songs of the week on StreetVoice streaming platform
- Enhanced output relevancy and faithfulness by **22%** with re-ranking & metadata filtering for 3,885-song vector database queries
- Streamlined app deployment by containerizing Qdrant, Ollama & Streamlit UI with Docker, enabling one-click deployment in 3 min

Imbalanced Classification Using Neyman-Pearson Paradigm and Cost-Sensitive Learning (Package link) **Oct. 2021-Apr. 2023**

- Fine-tuned Random Forest, XGBoost, LightGBM & CatBoost models for Alzheimer's disease prediction, achieved **26%** class-specific error rate reduction by implementing asymmetric error control for the highest performing CatBoost model (AUC=**0.88**)
- Increased feasibility of multi-class NP algorithm to **15.83x** by integrating the algorithm using R Caret machine learning framework
- Automated **91%** model resampling, visualization, and evaluation process using R ggplot, tidyverse, and published software on CRAN

Feature Tokenized Transformer Classifier with Metagenomic Profiles (GitHub link) **Jan. 2023-Apr. 2023**

- Developed Transformer model with PyTorch on HPC system to predict disease risk using microbiome profiles, achieving 0.85 F1 score
- Improved 7% balanced accuracy through regularization & customized deep learning layers using Ray Tune

EDUCATION

New York University, New York, NY **May 2023**

M.S. in Biostatistics

National Chiao Tung University, Hsinchu, Taiwan **Jan. 2019**

B.S. in Biological Science and Technology

SKILLS

Programming: Python (polars, pandas, numpy, scipy, seaborn, matplotlib), R (tidyverse, dplyr, ggplot2, MICE), Java, bash, SAS, Excel

Tools: AWS (S3, DDB, Redshift, GLUE, Athena, EC2, Lambda), Spark, High-performance computing, Git, SQL, Airflow, Docker, Tableau

ML/DL: PyTorch, OpenClaw, LangChain, LangGraph, LlamaFactory, Hugging face, SageMaker, Bedrock, Sklear, TensorFlow, Glmnet

PUBLICATIONS

[1] P. Wang, H. Liu, B. Yuan, **C. T. Tsai**, J. Fan, C. Kanar (2025) "Collaborative Large Language Model Agents for Ads Marketing Performance Diagnosis", *Amazon Machine Learning Conference*

[2] Y. Tian, **C. T. Tsai**, Y. Feng (2023) "npcs: Neyman-Pearson Classification via Cost-Sensitive Learning", *CRAN (link)*

[3] R. Garcia, **C. T. Tsai**, G. Ghosh, N. Jandhyala, Y. Feng, H. Rao, C. Thomas, B. Cohen, M. Karajannis, M. Snuderl, J. Allen, D. Segal (2024), "A single-center retrospective study of midbrain gliomas: clinically and radiographically heterogeneous tumors", *Neuro-Oncology Journal (under review)*